Requirement on academic honesty (COMP8090SEF/S209W):

1. Each student must make the following declaration on the cover page of the report

Hong Kong Metropolitan University

COMP S209W/8090SEF
Data Structures, Algorithms And Problem Solving
Data Structures, Algorithms

Course Project Report

Group 10

Submission date:

I declare that:
(i) I have read and checked that all parts of the project (including proposal, code programs, and reports), they are contributed by me, here submitted is original except for source material explicitly acknowledged;
(ii) the project, in parts or in whole, has not been submitted for more than one purpose without declaration;
(iii) I am aware of the University's policy and regulations on honesty in academic work and understand the possible consequence when breaching such policy and regulations;
(iv) I confirm that I have declared in the report about the usage of AI and other generative models, including but not limited to ChatGPT, LLaMA, Gemini, Mistral, and Stable Diffusion, and complied with the instructions provided by HKMU; and
(v) I am aware that I should be held responsible and liable to disciplinary actions, irrespective of whether I have signed the declaration and whether I have contributed, directly or indirectly, to the problematic contents.

I confirm that I have read through and understood the project requirements. I understand that failure to comply with the project requirements will result in score deduction.

NAME	SID
NGAN Chi Ho	11552520

2. Each student must acknowledge ALL external resources or code from others referenced. We will only mark the parts that are original, not the parts from others.

1.Introduction: This project presents the design and implementation of a “Music Lesson Management System” which intended for private music teachers. The motivation comes from my real-life experience. One day, I arrived at my private music teacher’s studio for a markup lesson, however, he forgot my lesson. He relied on memory and informal WhatsApp message to manage lesson appointments. As a result, the lesson could not take place as planned. 1.1 Problem: This situation are not only inconveniences, they can affect teaching efficiency, student satisfaction, financial tracking, and overall professionalism. A private tutor often performs several roles simultaneously, including instructor, planner, and accountant. Without a dedicated system, important information may be scattered across chat histories, handwritten notes, and payment records. Consequently, they can easily lead to missed appointments, duplicated bookings, weak financial record-keeping, or confusion when lessons are rescheduled. These issues are frequently discussed in online forums with same issue, where common solutions involve the use of commercial management systems or mobile applications, which may be costly or overly complex for individual tutors. 1.2 Solution: The aim of this project is to develop a lightweight but practical software system for managing students, lesson schedules, lesson changes, time conflict checking, invoice generation, and monthly income reporting. 2. Overview: This system starts with a simple login system in the CLI by running “main.py”, where the user enters the default account and password to access the program. After login, user may choose in GUI or CLI mode. The CLI is useful for direct text input and basic operations, while the GUI provides a more user-friendly and visual interface. In GUI mode, all student records are displayed in a table showing key information such as name, instrument, grade, lesson mode, lesson day, start time, lesson duration, fee, and place. This allows users to view and manage all data in one screen more efficiently instead of reading line by line from terminal output. The system supports core functions including adding, editing, and deleting students, generating invoices, exporting monthly income statements, and managing lessons such as cancellation, rescheduling, and conflict detection. In addition, the monthly schedule panel allows users to view the full timetable or check which students have lessons on a specific date. The system is divided into multiple modules. “models.py” defines the core domain entities, “repository.py” handles persistent storage, “services.py” coordinates student-related operations, “schedule_service.py” manages scheduling logic, “invoice.py” generates invoice, “validators.py” centralizes input validation, and the GUI and CLI files provide the user interface. The application is “bootstrapped” through a factory function that wires together the “repository”, “student service”, and “invoice”. 2.1. Core Function in Schedule Generation: For regular students, the system generates lesson dates in a month by following a fixed weekly pattern. For flexible students, the system reads explicitly stored custom lesson records and selects those that

fall within the requested month. These two behaviors are encapsulated in separate lesson plan classes, while the scheduling service decides which strategy to use in “schedule_service.py” that “get_plan()” returns either “RegularLessonPlan()” or “FlexibleLessonPlan()”. Depending on whether the student is regular or flexible. Both plan in “class LessonPlan()” and implement the same abstract interface in “lesson_plan.py”. 2.2 Lesson Change: A recurring weekly schedule is not fully realistic. For this reason, the project introduces lesson overrides. A lesson override stores a change to an originally scheduled regular lesson that may be changed, cancelled, or moved. Through “lesson_overrides”, which is stored inside the “Student” object as a dictionary. Each override maps an original regular lesson date to a changed lesson record. “RegularLessonPlan” uses this dictionary to determine whether a scheduled lesson should remain unchanged, appear as cancelled, or be moved to another date and time. The GUI includes buttons for “Change Selected Lesson,” “Cancel Selected Lesson,” 2.3 Conflict detection and validation: The system does not merely store data, it actively checks whether schedules are valid. The validation layer is centralized in “InputValidator”, which checks date format, time format, weekday, place, and lesson duration. On top of this, “validate_student_schedule()” checks whether a regular student’s weekly lesson conflicts with an existing timetable and whether a flexible student contains duplicate or overlapping lessons. “check_single_lesson_conflict()” further verifies that a changed lesson does not overlap with another concrete lesson on the same date. 2.4 Schedule Presentation: “schedule_for_month()” produce a full monthly schedule, allowing the user to inspect all active lessons within a specified month. Second, it can display which students have lessons on a specific date. The selected student preview panel provides a detailed view of the chosen student’s schedule for the current month which including any changed or cancelled lessons. It also supports sorting by different columns. 2.5 Invoice and Statement: An “Invoice” object for a selected student and month by collecting all active lesson and calculating the total fee accordingly. The “InvoiceService“ then exports in PDF. In addition, the system can produce a monthly income statement, giving the teacher a clearer earnings markdown. 3. Object Oriented programming: Encapsulation is demonstrated by grouping related data and operations within the same classes. The “Student” class stores all important student information in one structured object, including identity, lesson details, scheduling mode, overrides, and custom lessons. Likewise, “LessonOccurrence” encapsulates the details of an actual lesson session, and “Invoice” encapsulates billing-related data. 3.1 Abstraction: It is visible in both the domain and persistence layers. In “repository.py”, the abstract class “StudentRepository” defines “list_students()”, “get_by_id()”, “add()”, “update()”, “delete_by_id()”, and “next_id()” without fixing the storage mechanism. This allows the concrete “JsonStudentRepository” class to

implement the details while preserving a clear abstraction boundary. 3.2 Inheritance: It appears in the relationship between “Student” and “Person” in “models.py”, where “Student” provides the concrete implementation of “display_name()”. It also appears in the lesson planning structure, where multiple lesson-plan classes follow a common abstract lesson-plan interface. 3.3 Polymorphism: It is demonstrated by “get_plan()” in “schedule_service.py”. This method may return either a regular or flexible lesson-plan object, after which the program simply calls get_occurrences() on the result. The calling code does not need to know which subclass was selected, because both share the same interface. 3.4: Composition: In “bootstrap.py”, it integrates repository, service, and invoice components using dependency injection and composition, ensuring a loosely coupled and scalable system architecture. 3.5 Class Methods: The “from_dict()” in models.py uses @classmethod and reconstructs a “Student” instance from JSON dictionary data by returning. This is an appropriate design because the logic for rebuilding student objects. 3.6 Static Methods: the “InputValidator” utility in “validators.py”, which is referenced throughout the codebase from “cli.py”, “gui.py”, “services.py” and so on. Methods such as validate_name()do not depend on instance state for validation centralized. 3.7 Magic Methods: It mainly use in “gui.py” for display. 3.8 Class attributes: For defines most of the shared value like name, instrument, grade in class “Student” with different data type. 5. Library: Python built-in libraries such as “json” for data storage and retrieval, “datetime” for handling date and time operations, “pathlib” for file path management, and “Tkinter” for developing the graphical user interface, as well as a third-party library, “ReportLab”, which is used for generating PDF documents such as invoices and income statements. 6. Test case: Default regular student saves in JSON and manual added in the student list. Suppose Student A already has a lesson on Monday at 1300. If the user attempts to add Student B with the same date and time. It will reject the operation and raise a conflict error with overlapping time. The expected result is no record being added. When the lesson changes to another date. The expected result is that the selected lesson appears with [CHANGED] in the preview and the cancelled lesson which appears with [CANCELLED]. The calculations in lesson will not include it. 7. Weakness: The system uses a JSON repository, which is simple but inefficient for large data and concurrent use. It can be improved by using a database such as SQLite. Second, the authentication is default, it should be replaced with secure user login system. Also, it lacks a notification feature for reminders or schedule updates. All idea can be added for future improvement and expand to more similar self-employee like private teacher, sport instructor. 8 Conclusion: It addresses the key difficulties faced by private music teachers by combining student records, automated schedule, conflict detection, flexible-lesson handling, lesson override management, invoice generation, and income reporting.

Reference:

Python Student Management System – Simplify Your School Operations:

https://pythongeeks.org/python-student-managemen:t-system/

Student Results Management System Using Tkinter

https://www.geeksforgeeks.org/python/student-results-management-system-using-tkinter/

Appendix

GitHub link:

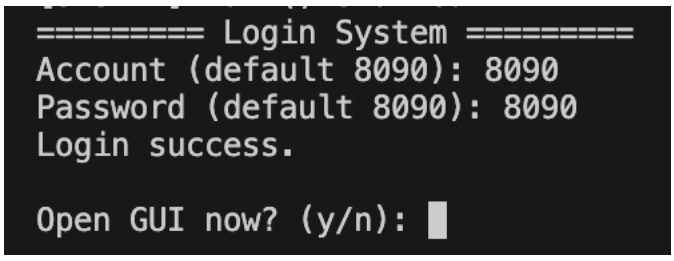
https://github.com/johnngan0511/11552520

1.Real forum discussion and solution

https://www.reddit.com/r/MusicTeachers/comments/1pj4xwr/music_teachers_what_software_do_you_use_to_manage/

https://www.reddit.com/r/smallbusiness/comments/1q4yxp4/software_recommendations_for_scheduling_in_music/?utm_source=chatgpt.com

2.Login In AC PW

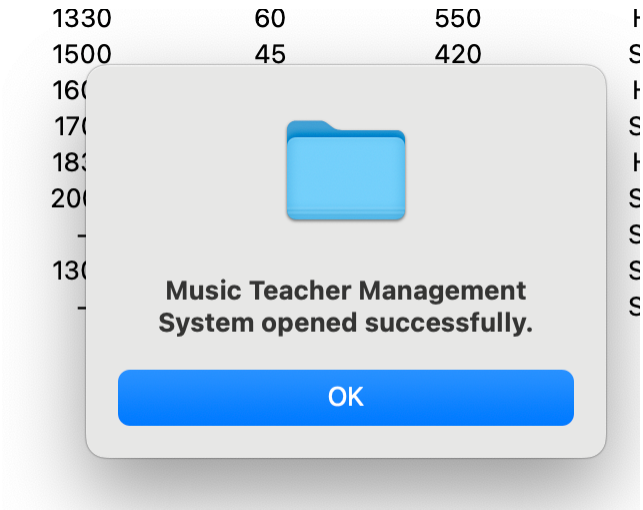


2. CLI Manu

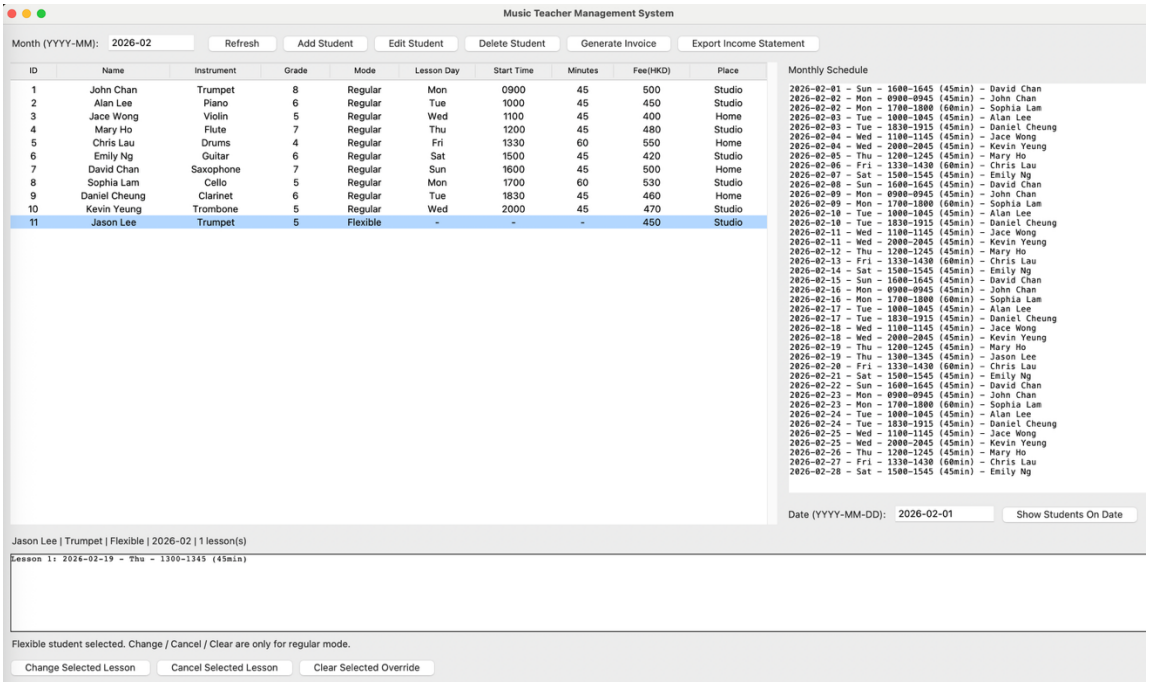
```
=== Music Teacher Management ===
1) View my students
2) Add student (regular / flexible)
3) Edit student
4) Delete student
5) Show my schedule (one month)
6) Generate invoice (PDF)
7) Add flexible lesson
8) Remove flexible lesson
0) Exit
Choose: 0
Bye!!!
[SYSTEM] main() finished
```

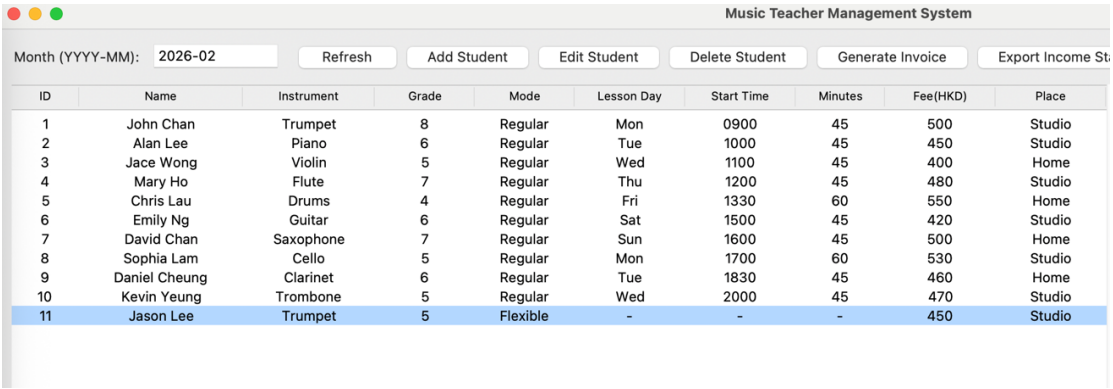
```
Choose: 1
=== My Students ===
ID=1 | Name=John Chan | Birth=2001-01-01 | Instrument=Trumpet | Grade=8 | Mode=Regular | Start=0900 | Time=45min | Fee=HKD500 | Lesson=Mon | Place=Studio
ID=2 | Name=Alan Lee | Birth=2002-03-12 | Instrument=Piano | Grade=6 | Mode=Regular | Start=1000 | Time=45min | Fee=HKD450 | Lesson=Tue | Place=Studio
ID=3 | Name=Jace Wong | Birth=2003-07-20 | Instrument=Violin | Grade=5 | Mode=Regular | Start=1100 | Time=45min | Fee=HKD400 | Lesson=Wed | Place=Home
ID=4 | Name=Mary Ho | Birth=2000-11-02 | Instrument=Flute | Grade=7 | Mode=Regular | Start=1200 | Time=45min | Fee=HKD480 | Lesson=Thu | Place=Studio
ID=5 | Name=Chris Lau | Birth=2004-09-14 | Instrument=Drums | Grade=4 | Mode=Regular | Start=1330 | Time=60min | Fee=HKD550 | Lesson=Fri | Place=Home
ID=6 | Name=Emily Ng | Birth=2005-05-10 | Instrument=Guitar | Grade=6 | Mode=Regular | Start=1500 | Time=45min | Fee=HKD420 | Lesson=Sat | Place=Studio
ID=7 | Name=David Chan | Birth=2002-12-25 | Instrument=Saxophone | Grade=7 | Mode=Regular | Start=1600 | Time=45min | Fee=HKD500 | Lesson=Sun | Place=Home
ID=8 | Name=Sophia Lam | Birth=2003-08-18 | Instrument=Cello | Grade=5 | Mode=Regular | Start=1700 | Time=60min | Fee=HKD530 | Lesson=Mon | Place=Studio
ID=9 | Name=Daniel Cheung | Birth=2001-04-30 | Instrument=Clarinet | Grade=6 | Mode=Regular | Start=1830 | Time=45min | Fee=HKD460 | Lesson=Tue | Place=Home
ID=10 | Name=Kevin Yeung | Birth=2004-06-22 | Instrument=Trombone | Grade=5 | Mode=Regular | Start=2000 | Time=45min | Fee=HKD470 | Lesson=Wed | Place=Studio
ID=11 | Name=Jason Lee | Birth=2001-01-01 | Instrument=Trumpet | Grade=5 | Mode=Flexible | Start=- | Time=- | Fee=HKD450 | Lesson=- | Place=Studio
ID=12 | Name=Jason Wong | Birth=2001-01-01 | Instrument=Trumpet | Grade=6 | Mode=Regular | Start=1300 | Time=45min | Fee=HKD500 | Lesson=Wed | Place=Studio
ID=13 | Name=Peter Lee | Birth=2001-01-01 | Instrument=Trumpet | Grade=5 | Mode=Flexible | Start=- | Time=- | Fee=HKD500 | Lesson=- | Place=Studio
```

2.GUI interface open

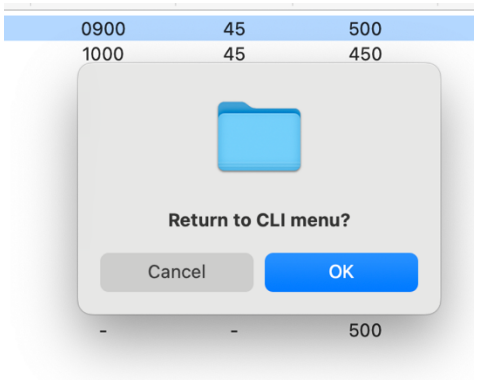


2.GUI interface



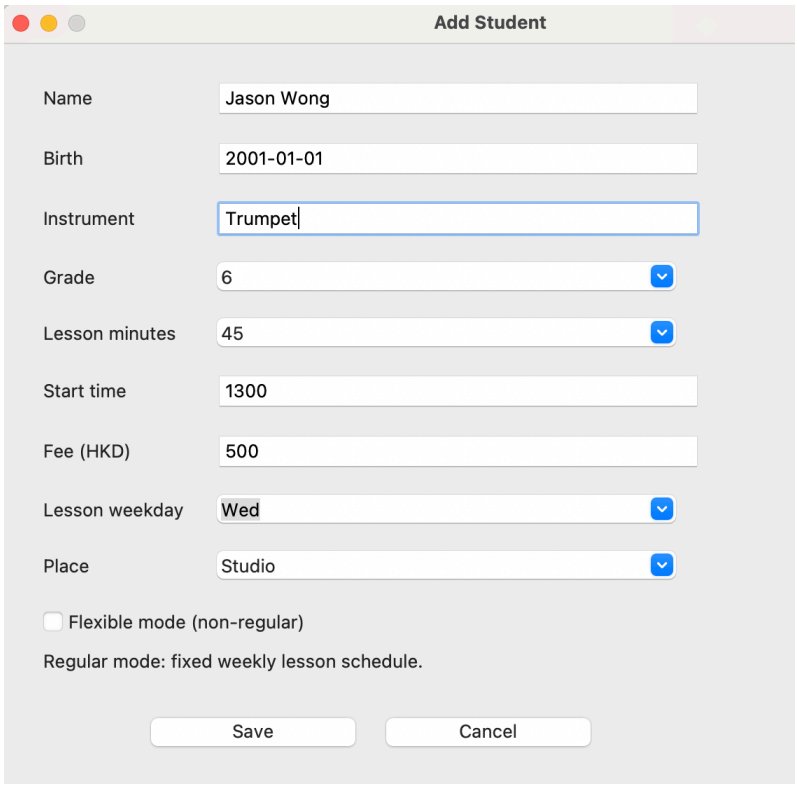


2. GUI close

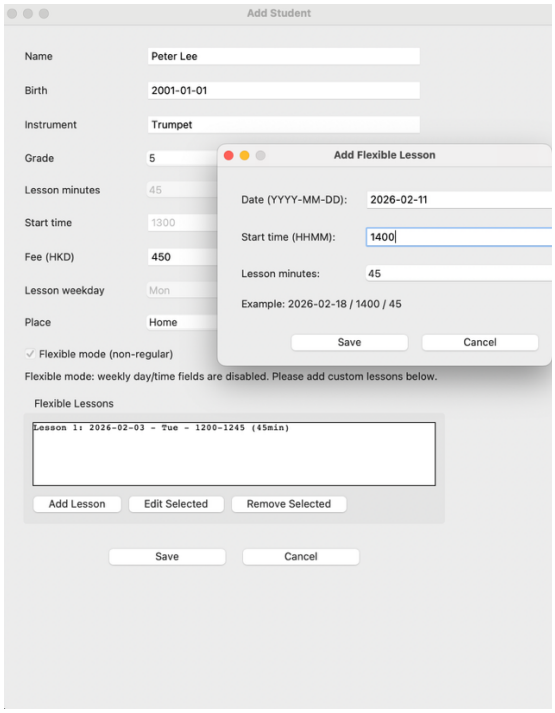



```
=== Music Teacher Management ===
1) View my students
2) Add student (regular / flexible)
3) Edit student
4) Delete student
5) Show my schedule (one month)
6) Generate invoice (PDF)
7) Add flexible lesson
8) Remove flexible lesson
0) Exit
Choose: 0
Bye!!!
[SYSTEM] main() finished
```

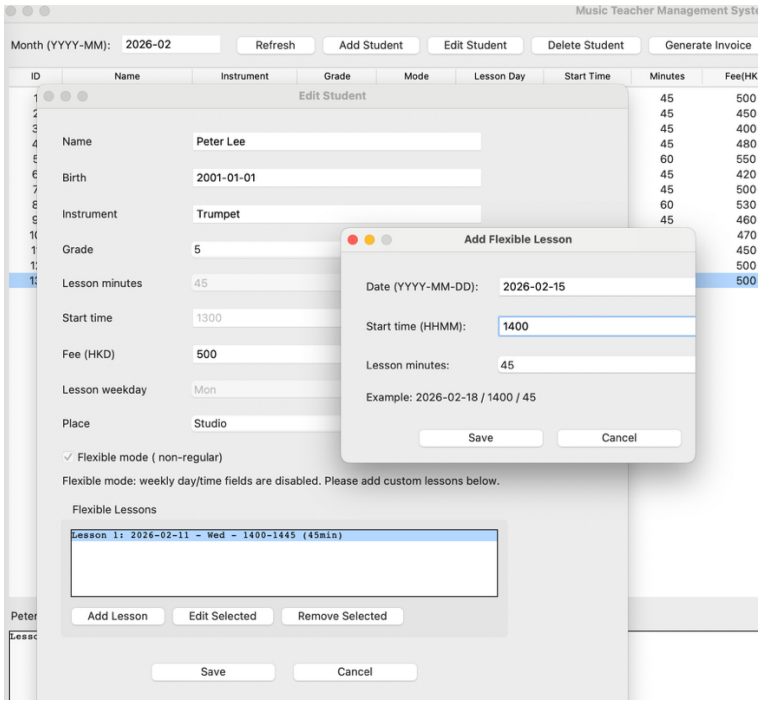
2.1 Student add in Regular



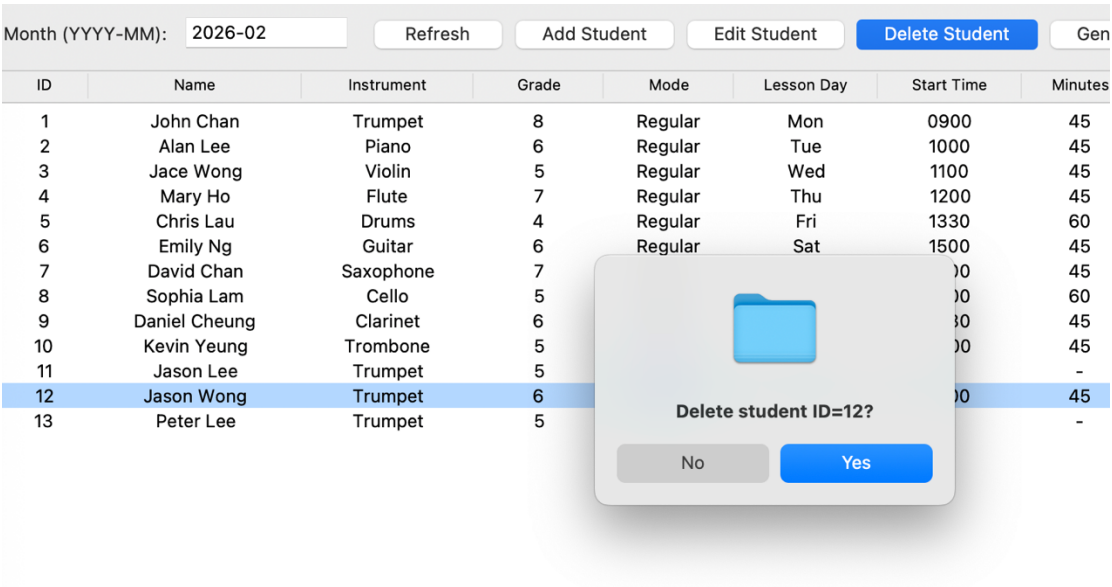
2.2 Student add in Flexible



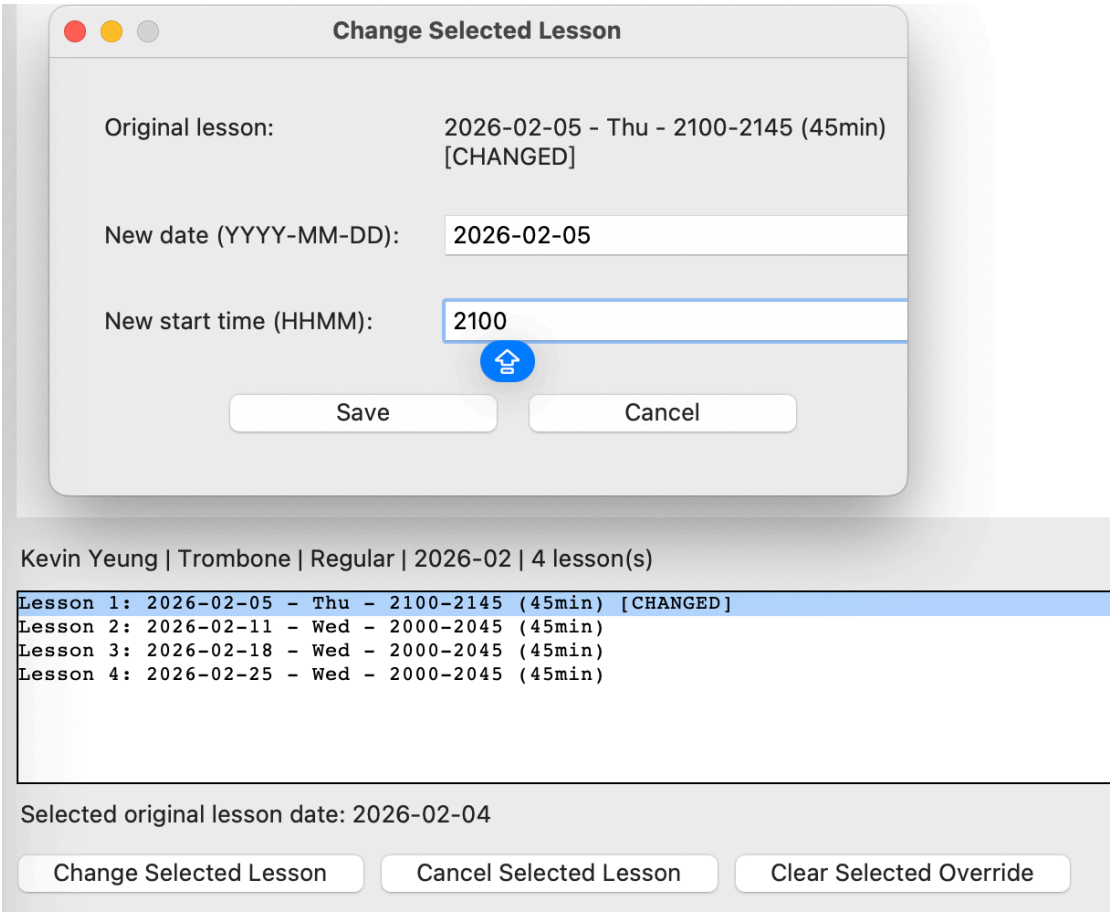
2.2 Edit

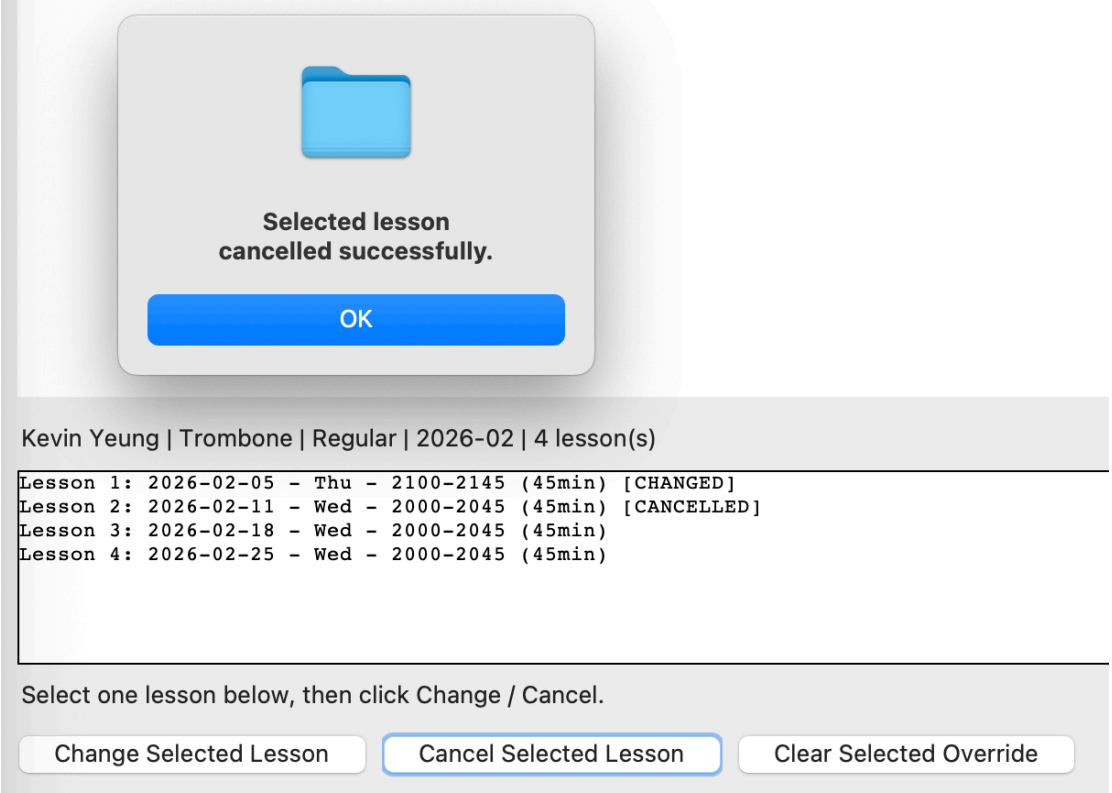


2.2 Delete Student

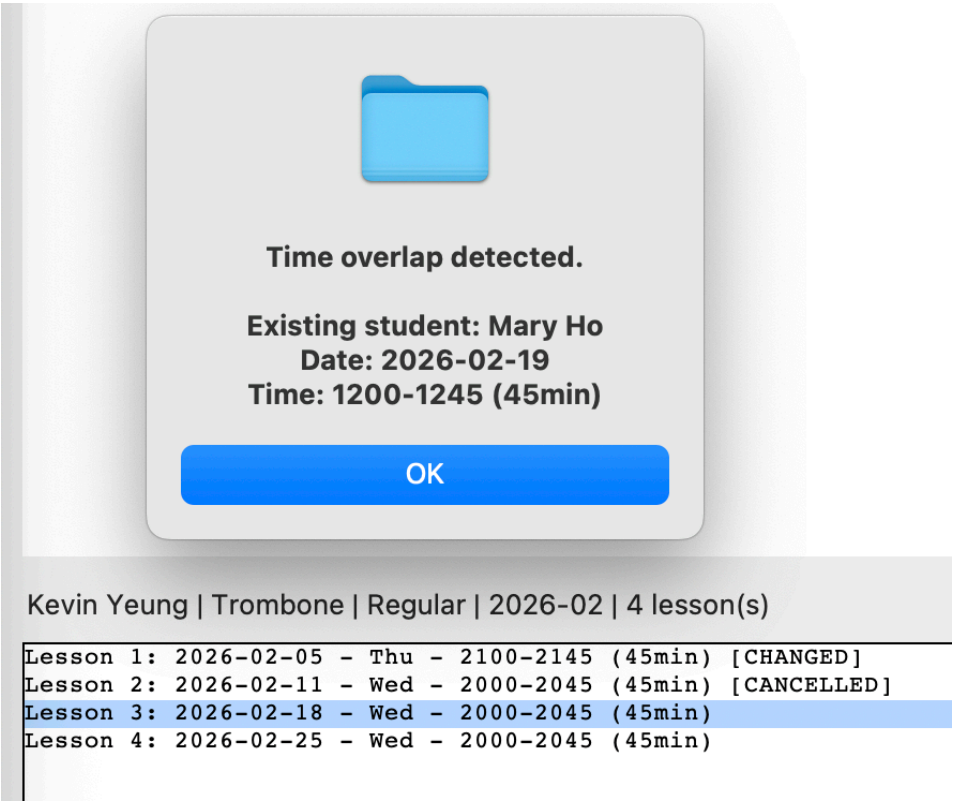


2.2 Lesson changed, cancelled and button




Selected lesson
cancelled successfully.
OK
Kevin Yeung | Trombone | Regular | 2026-02 | 4 lesson(s)
Lesson 1: 2026-02-05 - Thu - 2100-2145 (45min) [CHANGED]
Lesson 2: 2026-02-11 - Wed - 2000-2045 (45min) [CANCELLED]
Lesson 3: 2026-02-18 - Wed - 2000-2045 (45min)
Lesson 4: 2026-02-25 - Wed - 2000-2045 (45min)
Select one lesson below, then click Change / Cancel.
Change Selected Lesson
Cancel Selected Lesson
Clear Selected Override

2.3 Overlap


Time overlap detected.
Existing student: Mary Ho
Date: 2026-02-19
Time: 1200-1245 (45min)
OK
Kevin Yeung | Trombone | Regular | 2026-02 | 4 lesson(s)
Lesson 1: 2026-02-05 - Thu - 2100-2145 (45min) [CHANGED]
Lesson 2: 2026-02-11 - Wed - 2000-2045 (45min) [CANCELLED]
Lesson 3: 2026-02-18 - Wed - 2000-2045 (45min)
Lesson 4: 2026-02-25 - Wed - 2000-2045 (45min)

2.4 Schedule In Day

```
Students on 2026-02-02
========================================
2026-02-02 - Mon - 0900-0945 (45min) - John Chan - Trumpet - Studio
2026-02-02 - Mon - 1700-1800 (60min) - Sophia Lam - Cello - Studio
```

Date (YYYY-MM-DD): 2026-02-02 Show Students On Date

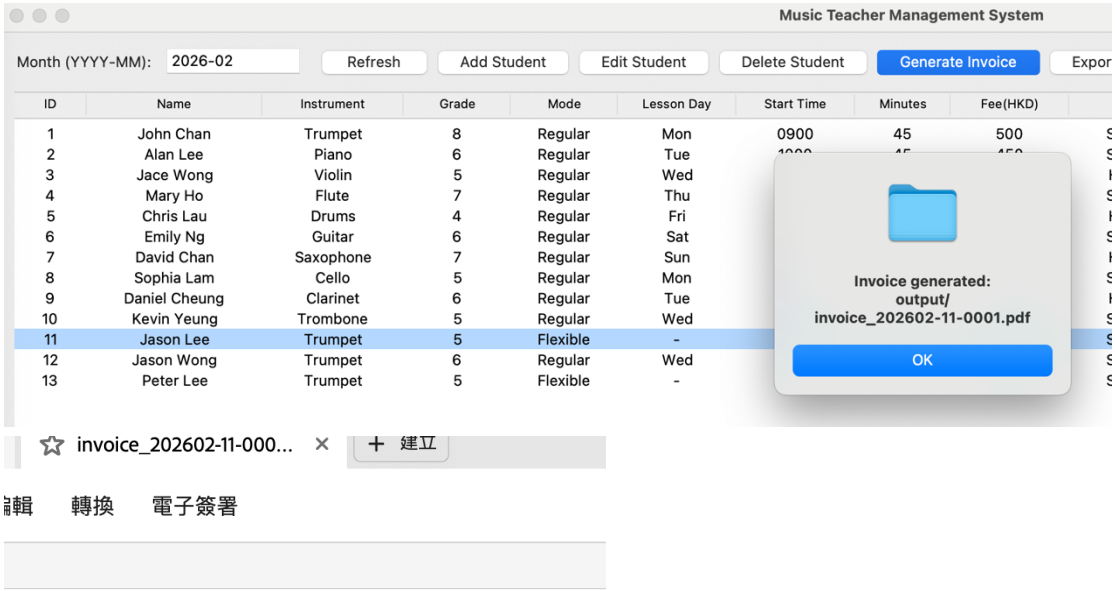
2.4 Schedule In Month

Monthly Schedule

```
2026-02-01 - Sun - 1600-1645 (45min) - David Chan
2026-02-02 - Mon - 0900-0945 (45min) - John Chan
2026-02-02 - Mon - 1700-1800 (60min) - Sophia Lam
2026-02-03 - Tue - 1000-1045 (45min) - Alan Lee
2026-02-03 - Tue - 1830-1915 (45min) - Daniel Cheung
2026-02-04 - Wed - 1100-1145 (45min) - Jace Wong
2026-02-04 - Wed - 1300-1345 (45min) - Jason Wong
2026-02-05 - Thu - 1200-1245 (45min) - Mary Ho
2026-02-05 - Thu - 2100-2145 (45min) - Kevin Yeung
2026-02-06 - Fri - 1330-1430 (60min) - Chris Lau
2026-02-07 - Sat - 1500-1545 (45min) - Emily Ng
2026-02-08 - Sun - 1600-1645 (45min) - David Chan
2026-02-09 - Mon - 0900-0945 (45min) - John Chan
2026-02-09 - Mon - 1700-1800 (60min) - Sophia Lam
2026-02-10 - Tue - 1000-1045 (45min) - Alan Lee
2026-02-10 - Tue - 1830-1915 (45min) - Daniel Cheung
2026-02-11 - Wed - 1100-1145 (45min) - Jace Wong
2026-02-11 - Wed - 1300-1345 (45min) - Jason Wong
2026-02-11 - Wed - 1400-1445 (45min) - Peter Lee
2026-02-12 - Thu - 1200-1245 (45min) - Mary Ho
2026-02-13 - Fri - 1330-1430 (60min) - Chris Lau
2026-02-14 - Sat - 1500-1545 (45min) - Emily Ng
2026-02-15 - Sun - 1400-1445 (45min) - Peter Lee
2026-02-15 - Sun - 1600-1645 (45min) - David Chan
2026-02-16 - Mon - 0900-0945 (45min) - John Chan
2026-02-16 - Mon - 1700-1800 (60min) - Sophia Lam
2026-02-17 - Tue - 1000-1045 (45min) - Alan Lee
2026-02-17 - Tue - 1830-1915 (45min) - Daniel Cheung
2026-02-18 - Wed - 1100-1145 (45min) - Jace Wong
2026-02-18 - Wed - 1300-1345 (45min) - Jason Wong
2026-02-18 - Wed - 2000-2045 (45min) - Kevin Yeung
2026-02-19 - Thu - 1200-1245 (45min) - Mary Ho
2026-02-19 - Thu - 1300-1345 (45min) - Jason Lee
2026-02-20 - Fri - 1330-1430 (60min) - Chris Lau
2026-02-21 - Sat - 1500-1545 (45min) - Emily Ng
2026-02-22 - Sun - 1600-1645 (45min) - David Chan
2026-02-23 - Mon - 0900-0945 (45min) - John Chan
2026-02-23 - Mon - 1700-1800 (60min) - Sophia Lam
2026-02-24 - Tue - 1000-1045 (45min) - Alan Lee
2026-02-24 - Tue - 1830-1915 (45min) - Daniel Cheung
2026-02-25 - Wed - 1100-1145 (45min) - Jace Wong
2026-02-25 - Wed - 1300-1345 (45min) - Jason Wong
2026-02-25 - Wed - 2000-2045 (45min) - Kevin Yeung
```

Date (YYYY-MM-DD): 2026-02-01 Show Students On Date

2.5 Invoice



Music Lesson Invoice

Invoice No.: 202602-11-0001

Month: 2026-02

Student: Jason Lee

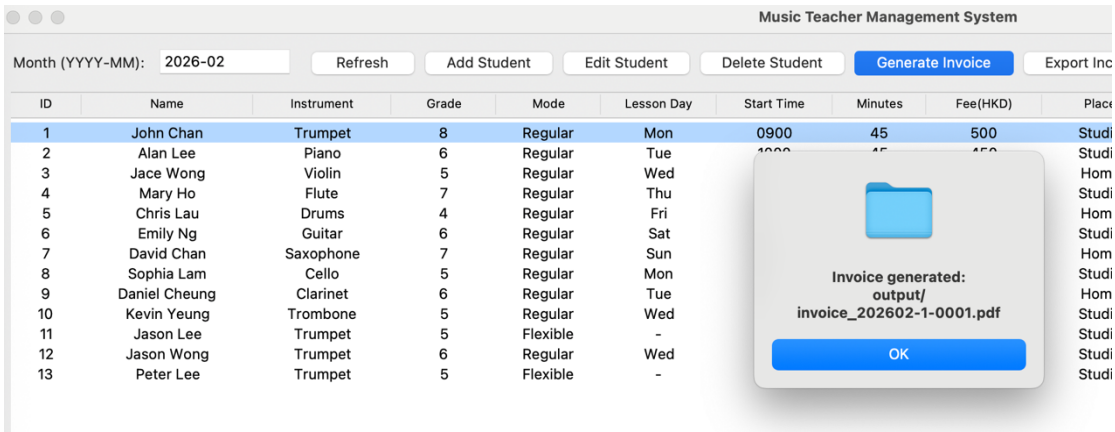
Instrument: Trumpet Grade: 5

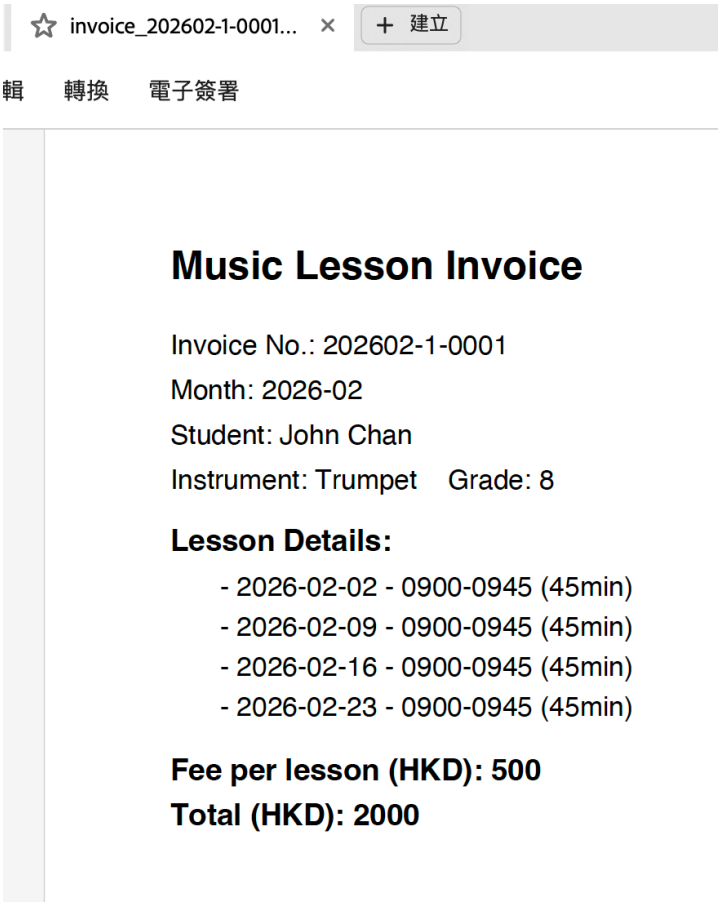
Lesson Details:

- 2026-02-19 - 1300-1345 (45min)

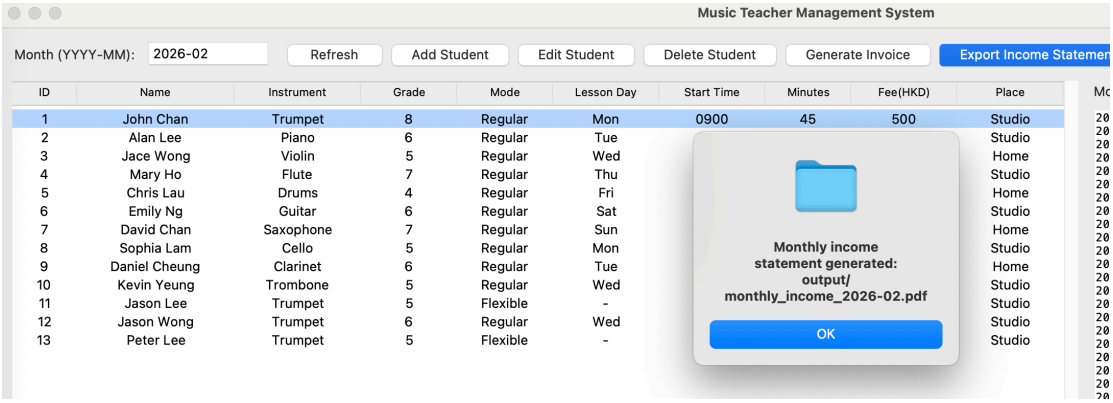
Fee per lesson (HKD): 450

Total (HKD): 450





2.5 Income statement



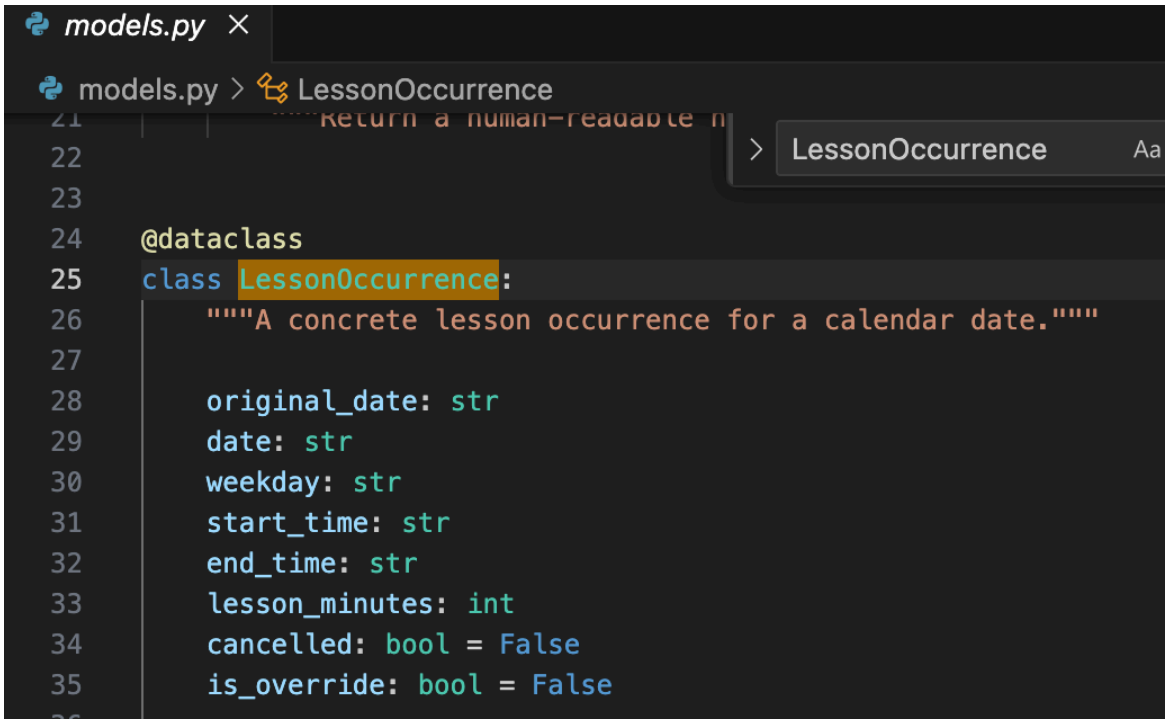
Monthly Income Statement

Month: 2026-02

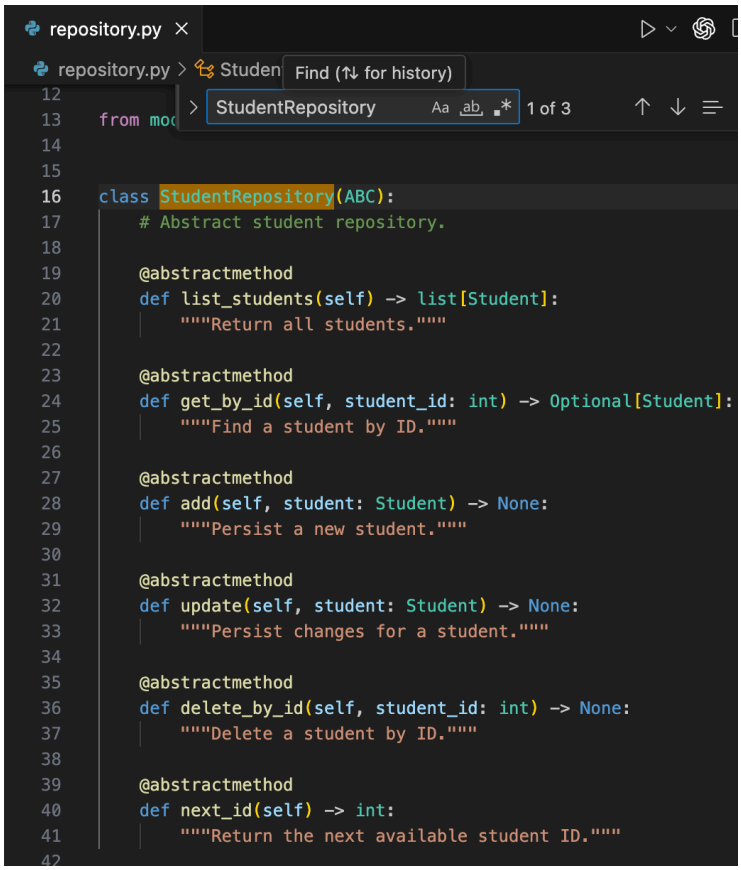
ID	Name	Instrument	Grade	Lessons	Fee	Total
1	John Chan	Trumpet	8	4	HKD 500	HKD 2000
2	Alan Lee	Piano	6	4	HKD 450	HKD 1800
3	Jace Wong	Violin	5	4	HKD 400	HKD 1600
4	Mary Ho	Flute	7	4	HKD 480	HKD 1920
5	Chris Lau	Drums	4	4	HKD 550	HKD 2200
6	Emily Ng	Guitar	6	4	HKD 420	HKD 1680
7	David Chan	Saxophone	7	4	HKD 500	HKD 2000
8	Sophia Lam	Cello	5	4	HKD 530	HKD 2120
9	Daniel Cheung	Clarinet	6	4	HKD 460	HKD 1840
10	Kevin Yeung	Trombone	5	3	HKD 470	HKD 1410
11	Jason Lee	Trumpet	5	1	HKD 450	HKD 450
12	Jason Wong	Trumpet	6	4	HKD 500	HKD 2000
13	Peter Lee	Trumpet	5	2	HKD 500	HKD 1000

Total Monthly Income: HKD 22020

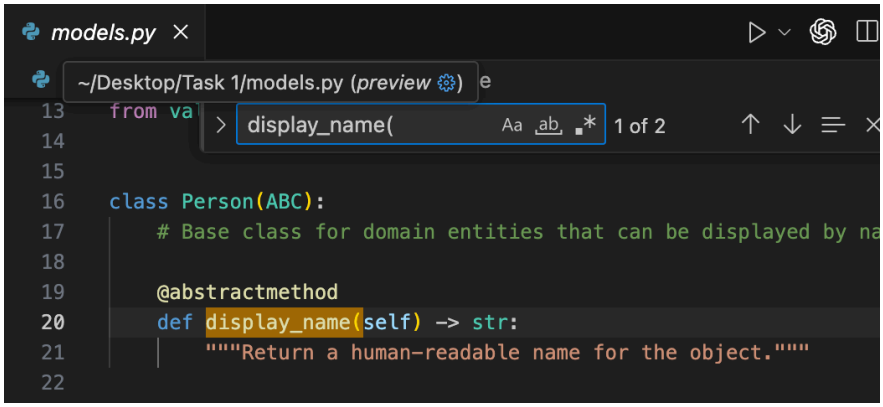
3. Encapsulation



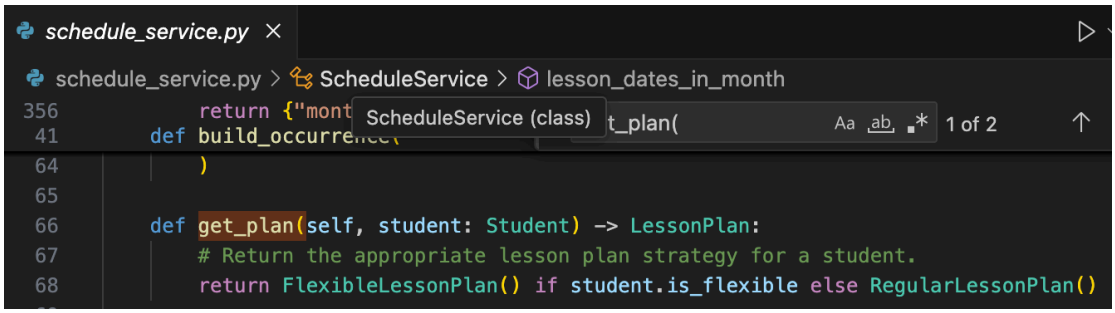
3.1 Abstraction

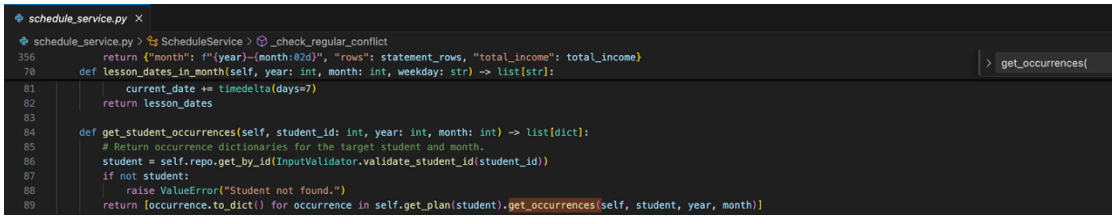


3.2 Inheritance

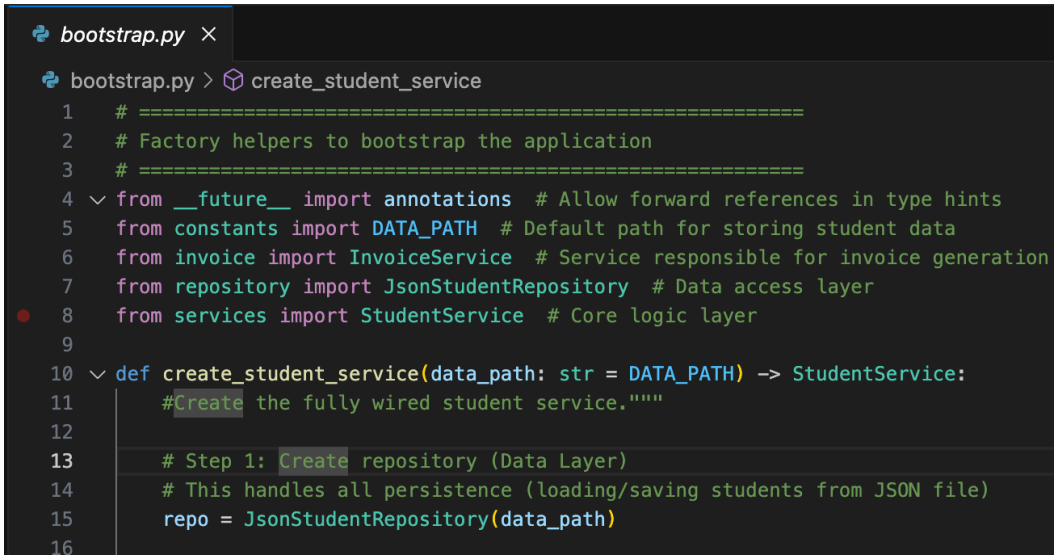


3.3 Polymorphism





3.4: Composition



3.5 Class Methods

3.6 Static Methods

validators.py

validators.py > InputValidator > validate_name

InputValidator Aa ab .* 2 of 15

```python
class InputValidator:
    """Centralized validation to keep UI code small and consistent."""

    @staticmethod
    def validate_non_empty_text(value: str, field_name: str) -> str:
        """Validate a required text field."""
        normalized_value = str(value).strip()
        if not normalized_value:
            raise ValidationError(f"{field_name} cannot be empty.")
        return normalized_value
```

3.7 Magic Methods

gui.py

gui.py > App

__init__ Aa ab .* 1 of 2

```python
class App(tk.Tk):

    def __init__(self, student_service):
        super().__init__()
        self.service = student_service
        self.title("Music Teacher Management System")
        self.geometry("1600x900")
        self.minsize(1200, 820)
        self.current_occurrences: list[dict] = []
        self.columns = ["id", "name", "instrument", "grade", "mode", "weekday"
        self.sort_states = {column_name: False for column_name in self.columns
        self.weekday_order = {name: index for index, name in enumerate(["Mon",
        self.grade_order = {grade: index for index, grade in enumerate(["1", "
        self._build_layout()
        self.refresh()
        self.show_schedule()
        self.after(200, self._show_open_message)  # show popup after GUI loads
        self.protocol("WM_DELETE_WINDOW", self.on_close)
```